Griffiths 2021 Breast Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony on sample (θ sweep 2/10/20; per-neighbourhood iLISI 1.7→19.2 at θ=10, the peak). The regenerated embedding is one continuous, thoroughly sample- and patient-mixed body; cell cycle is flat (no proliferation lobe). Only a 328-cell iCell8 technology streak + tiny specks detach (island removal 110,195→109,353). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

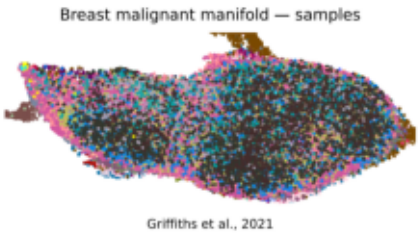
Organised Gast↔pSom cross-repression along one fold-free axis (pseudotime ρ: Gast +0.40, pSom −0.36) with a terminal Crest gradient (ρ +0.28) — the non-branching pattern. FLAG: Crest enriches the Gast-leaning terminus, not pSom (an inversion; passes anyway).

Gate 3 — hub cross-repression → NA

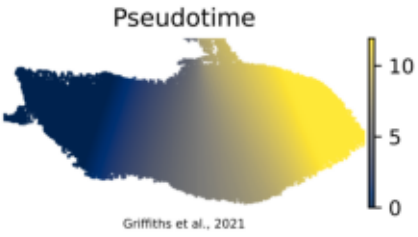
No bifurcation: a PPT tree yields 15 tips (13 branch nodes) all with identical module identity, and Gast/pSom co-express below the independence expectation (7.3% vs 9%) — no convergence hub. Gate 3 is branching-only, so NA.

STEP FIGURES

Manifold — samples



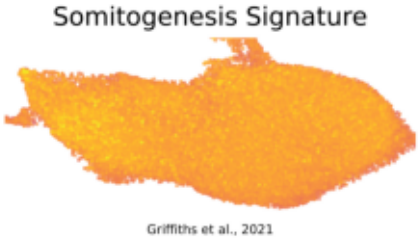
Pseudotime



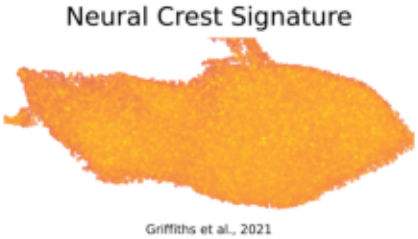
Gastrulation



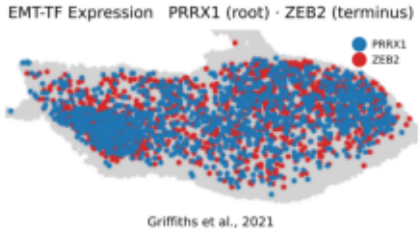
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (ER+ neoadjuvant letrozole ± ribociclib) (Griffiths et al., 2021), 10x + iCell8 — 109,353 malignant cells, 67 samples.

Verdict.
The embryonic programme reproduces as a COLLAPSED bifurcation: a single continuous gastrulation/somitogenesis ↔ neural-crest axis (geometry class Linear-pSom/Crest; module means Gast −0.019 / pSom −0.053 / Crest +0.023). Gast and pSom cross-repress along the axis but never resolve into two competing arms, so there is no hub and no Gate-3 test. Crest is terminal but enriches the Gast side (flag). EMT-TFs near-silent in this treated luminal cohort; the most detectable pseudotime-graded pair is PRRX1 (root) ↔ ZEB2 (terminus). Gate 1 GO / Gate 2 GO / Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 cells/sample (gap-finder degenerate on smooth dist) → 67/95 samples	**HVG / n_pcs**	4,072 over-dispersed genes; n_pcs = 20 (scree elbow ~PC7, ~32% cum. var)
Integration	Harmony θ = 10 (iLISI peak 19.2; θ=20 over-integrates)	**Island removal**	6×-median radius graph, largest component (842 cells / 57 specks removed)
Geometry	Non-branching fold-free curve (fold ratio 2.95); root = Gast/pSom pole	**Root / EMT-TFs**	pSom-high developmental origin; PRRX1 (early) ↔ ZEB2 (late)